



Post-Graduate Diploma in ML/AI

Course : Machine Learning

Lecture On : Pre exam session

Instructor : Harish Kumar

Modules Included

- Advanced Regression ✓
- Model Selection ✓
- Tree Models ✓
- Boosting ✓
- Clustering ✓

Overfitting

Training Accuracy $\rightarrow$ very high {90%}

Test Accuracy $\rightarrow$ very low {10%}

$\rightarrow$ Model is very Complex

Data is very less

Data not fit

Underfitting

Training Acc. $\rightarrow$ very low 60%

Test Acc $\rightarrow$ very low 60%

Reason:-

Model is very simple

cris Data is very large.

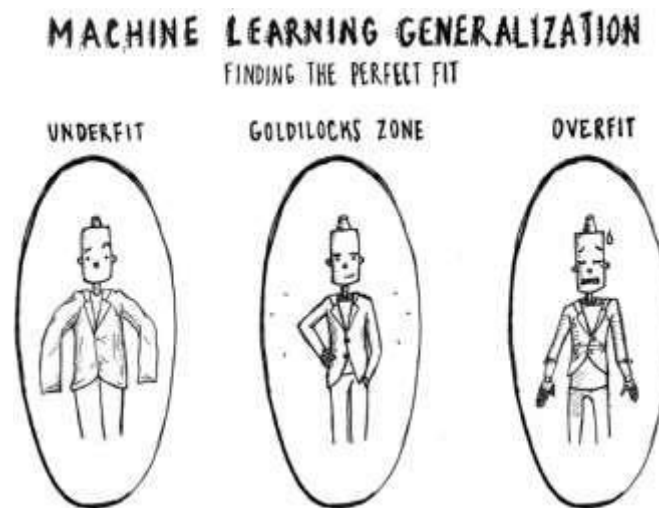
Ideal Model. upGrad

Training Acc. - Test acc. both are comparable

Ideal. $\rightarrow$ limit dot

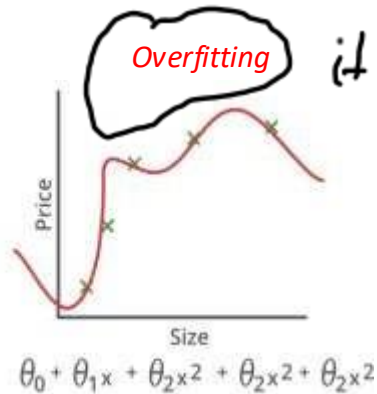
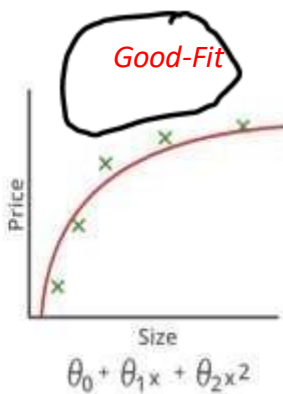
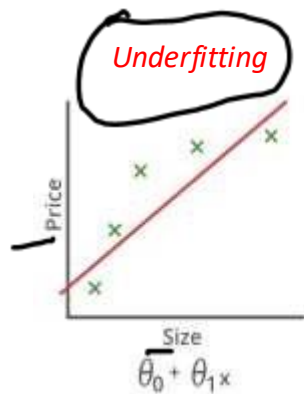
Underfitting refers to a model that can neither model the training data nor generalize to new data. An underfit machine learning model is not a suitable model and will be obvious as it will have poor performance on the training data.

Overfitting refers to a model that models the training data too well. Overfitting happens when a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new data. This means that the noise or random fluctuations in the training data is picked up and learned as concepts by the model. The problem is that these concepts do not apply to new data and negatively impact the model's ability to generalize.



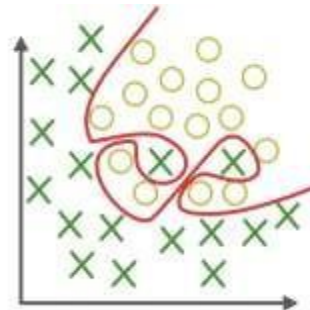
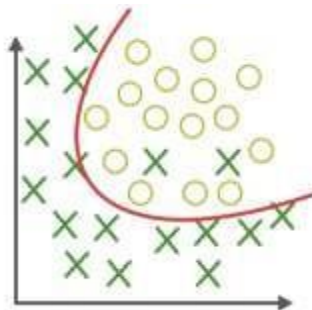
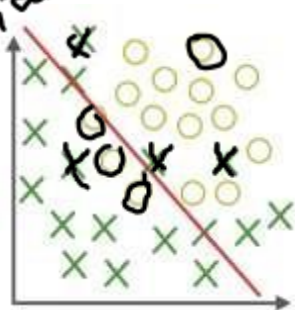
What & Why of Overfitting and Underfitting

Regression



it remembers data instead of creating from it.

Classification

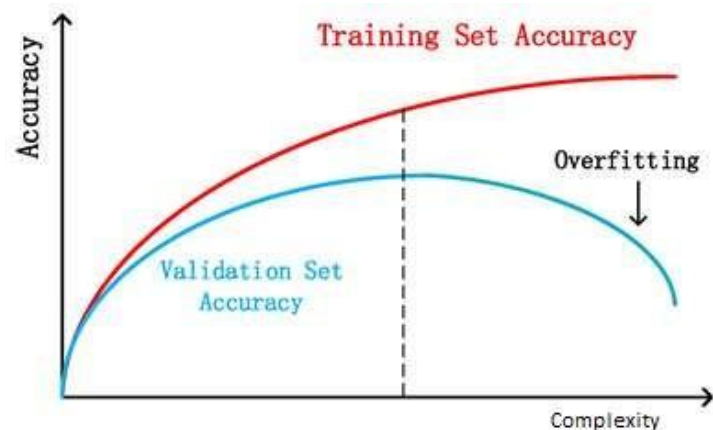


Detecting overfitting is almost impossible before you test the data. It can help address the inherent characteristic of overfitting, which is the inability to generalize data sets. The data can, therefore, be separated into different subsets to make it easy for training and testing. The data is split into three main parts, i.e., a train set, a test set & a validate set

- **Training set:** A set of examples used for learning, that is to fit the parameters of the classifier.

- **Validation set:** A set of examples used to tune the parameters of a classifier, for example to choose the number of hidden units in a neural network.

- **Test set:** A set of examples used only to assess the performance of a fully-specified classifier.



By segmenting the dataset, we can examine the performance of the model on each set of data to spot overfitting when it occurs. The performance can be measured using the percentage of accuracy observed in train & validate data sets to conclude on the presence of overfitting.

Q1badd Title

How does number of observations influence overfitting? Choose the correct answer(s)

1. In case of fewer observations, it is easy to overfit the data.
2. In case of fewer observations, it is hard to overfit the data.
3. In case of more observations, it is easy to overfit the data.
4. In case of more observations, it is hard to overfit the data.

A. 1 and 4

B. 2 and 3

C. 1 and 3

D. None of these



→ Practice in teams of 4 students

→ Industry expert mentoring to learn better

→ Get personalised feedback for improvements

Qbank1badd Title

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C. 1 and 3

D. None of these



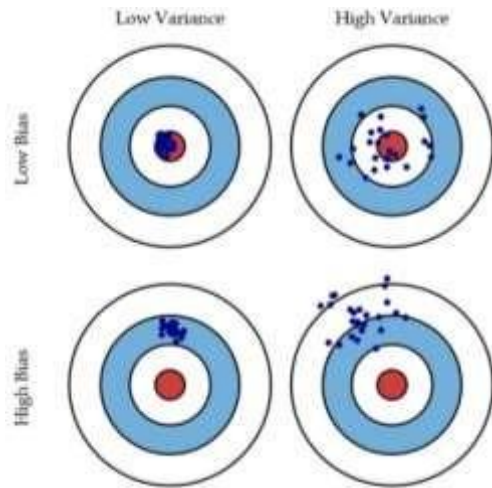
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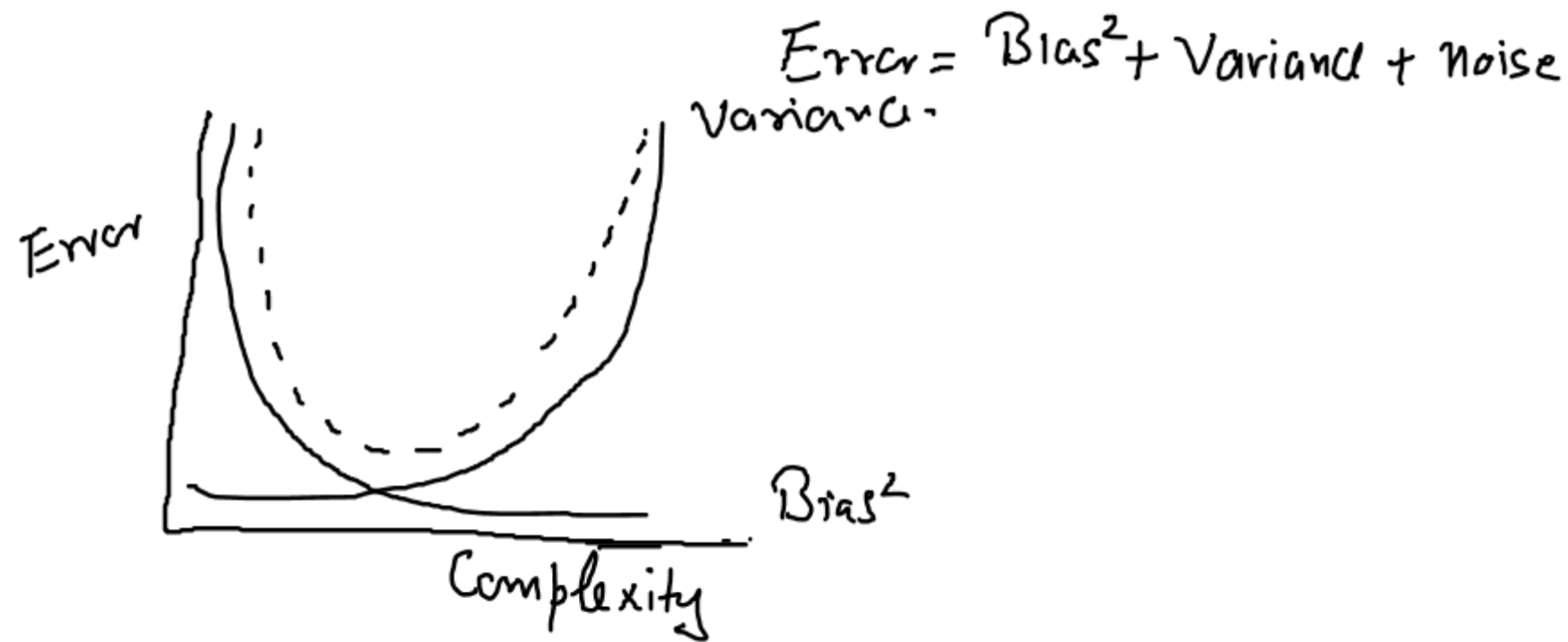
The prediction error for any machine learning algorithm can be broken down into three parts:

- Bias Error
- Variance Error
- Irreducible Error

The irreducible error cannot be reduced regardless of what algorithm is used. It is the error introduced from the chosen framing of the problem and may be caused by factors like unknown variables that influence the mapping of the input variables to the output variable.

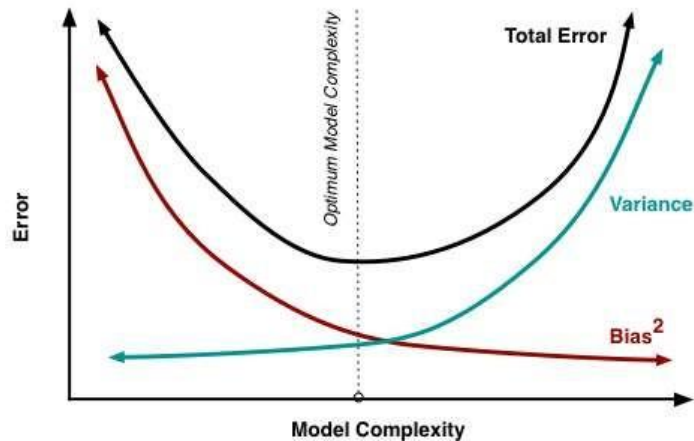
- **Low Bias:** Suggests less assumptions about the form of the target function.
- **High-Bias:** Suggests more assumptions about the form of the target function.
- **Low Variance:** Suggests small changes to the estimate of the target function with changes to the training dataset.
- **High Variance:** Suggests large changes to the estimate of the target function with changes to the training dataset.





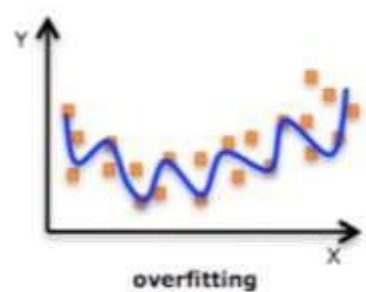
The goal of any supervised machine learning algorithm is to achieve low bias and low variance. In turn the algorithm should achieve good prediction performance.

- Parametric or linear machine learning algorithms often have a high bias but a low variance.
- Non-parametric or non-linear machine learning algorithms often have a low bias but a high variance.



The parameterization of machine learning algorithms is often a battle to balance out bias and variance.

One of the major aspects of training your machine learning model is avoiding overfitting. *The model will have a low accuracy if it is overfitting.* This happens because your model is trying too hard to capture the noise in your training dataset.



Regularization

This is a form of regression, that constrains/ regularizes or shrinks the coefficient estimates towards zero. In other words, ***this technique discourages learning a more complex or flexible model, so as to avoid the risk of overfitting.***

Regularization:- technique to penalize model for being overfitted.

LASSO (L1)

it perform Absolute Shrinkage

it makes less imp feature's

i) Coefficient equal to zero

$$ii) y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$iii) \text{Objective} \Rightarrow \text{Loss func} + \lambda \sum_{i=1}^n |\beta_i|$$

it can be use for feature selection.

Ridge (L2)

it makes less imp feature's coefficient close to zero

$$ii) y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Objective function =

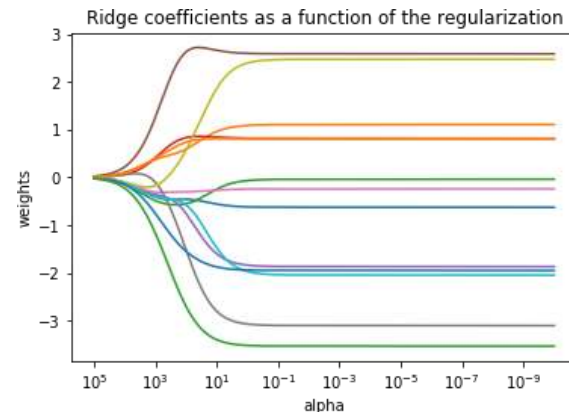
$$\text{Loss function} + \lambda \sum_{i=1}^n \beta_i^2$$

it can't be used for

RSS is modified by adding the shrinkage quantity.

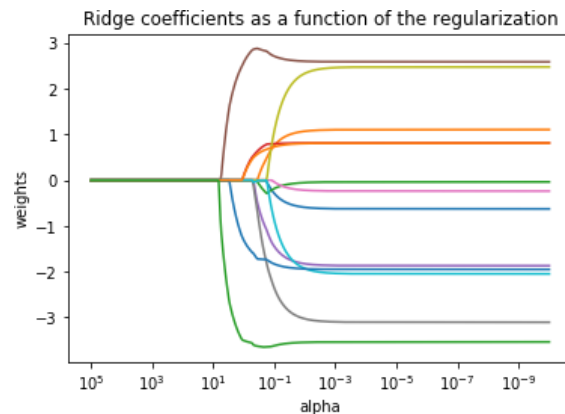
Now, the coefficients are estimated by minimizing this function. Here, **λ is the tuning parameter that decides how much we want to penalize the flexibility of our model.**

The increase in flexibility of a model is represented by increase in its coefficients, and if we want to minimize the above function, then these coefficients need to be small.



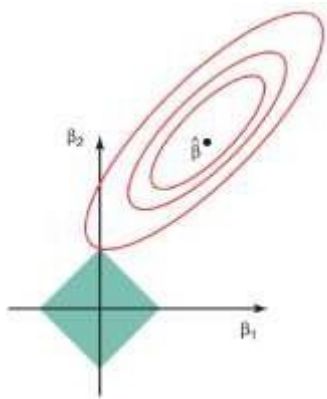
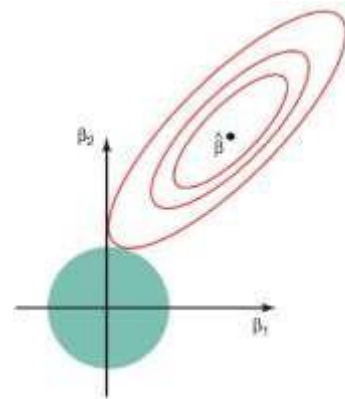
$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2$$

Lasso is another variation, in which the above function is minimized. Its clear that *this variation differs from ridge regression only in penalizing the high coefficients.*



$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| = \text{RSS} + \lambda \sum_{j=1}^p |\beta_j|.$$

The **ridge regression** is expressed by $\beta_1^2 + \beta_2^2 \leq s$. This implies that *ridge regression coefficients have the smallest RSS(loss function) for all points that lie within the circle given by $\beta_1^2 + \beta_2^2 \leq s$*



The lasso, the equation becomes, $|\beta_1| + |\beta_2| \leq s$. This implies that *lasso coefficients have the smallest RSS(loss function) for all points that lie within the diamond given by $|\beta_1| + |\beta_2| \leq s$.*

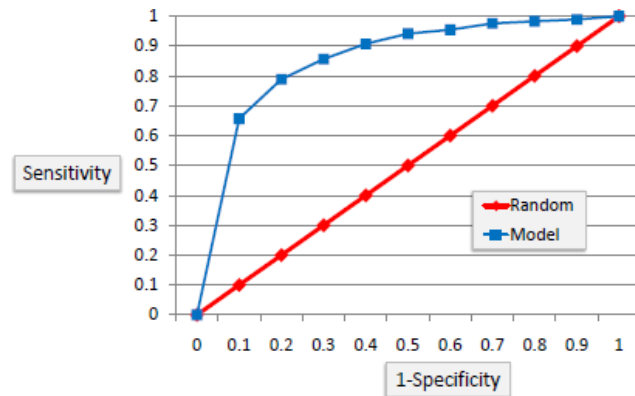
A confusion matrix shows the number of correct and incorrect predictions made by the classification model compared to the actual outcomes (target value) in the data. The matrix is $N \times N$, where N is the number of target values (classes). Performance of such models is commonly evaluated using the data in the matrix. The following table displays a 2x2 confusion matrix for two classes (Positive and Negative).

Confusion Matrix	Target			
	Positive	Negative		
Positive	a	b	Positive Predictive Value	$a/(a+b)$
Negative	c	d	Negative Predictive Value	$d/(c+d)$
	Sensitivity	Specificity	Accuracy = $(a+d)/(a+b+c+d)$	

- **Accuracy** : the proportion of the total number of predictions that were correct.
- **Positive Predictive Value** or **Precision** : the proportion of positive cases that were correctly identified.
- **Negative Predictive Value** : the proportion of negative cases that were correctly identified.
- **Sensitivity** or **Recall** : the proportion of actual positive cases which are correctly identified.
- **Specificity** : the proportion of actual negative cases which are correctly identified.

ROC Curves summarize the trade-off between the true positive rate and false positive rate for a predictive model using different probability thresholds

It's often used as a proxy for the trade-off between the sensitivity of the model (true positives) vs the fall-out or the probability it will trigger a false alarm (false positives).



$$\text{TPR / Recall / Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

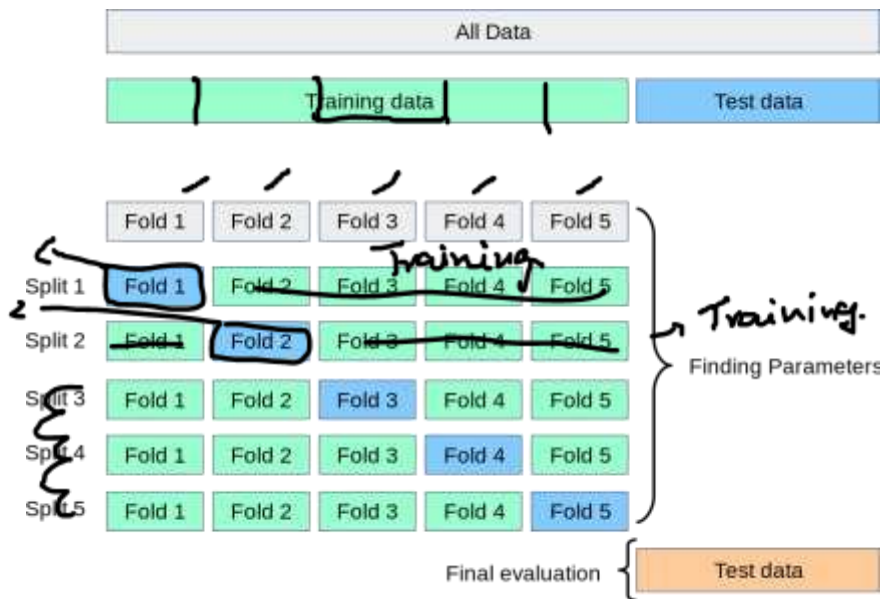
$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

$$F - \text{measure} = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}}$$

https://en.wikipedia.org/wiki/Confusion_matrix

Cross-validation is a resampling technique used to evaluate machine learning models on a limited data set.

The most common use of cross-validation is the k-fold cross-validation method. Our training set is split into K partitions, the model is trained on $K-1$ partitions and the test error is predicted and computed on the K^{th} partition. This is repeated for each unique group and the test errors are averaged across.

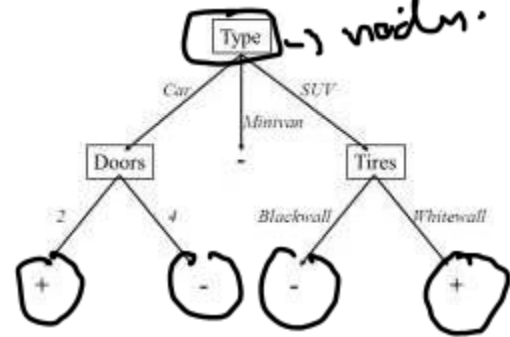


Decision Tree :- It is helpful in model Interpretability.

→ It creates Tree like Structure to explain problem.

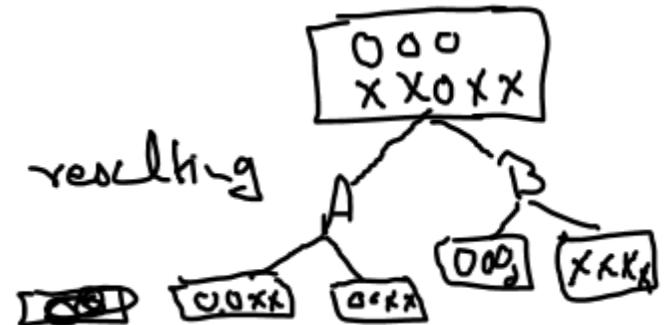
Decision tree is the most powerful and popular tool for classification and prediction. A Decision tree is a flowchart like tree structure, where each internal node denotes a test on an attribute, each branch represents an outcome of the test, and each leaf node (terminal node) holds a class label.

Type, Doors, Tires. 20 cl. =



Concept of homogeneity.

Choose Column as root which makes resulting nodes more homogeneous



to Calculate Homogeneity.

Impurity Criterion

Gini Index

$$\text{Gini} = \sum_{i=1}^k p_i^2$$

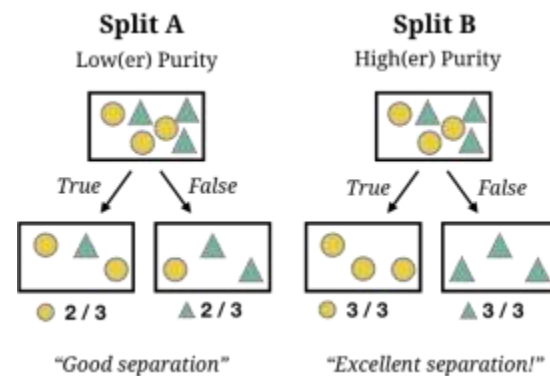
p_i : proportion of the samples that belongs to class c for a particular node

Entropy

$$\mathcal{E}[D] = -\sum_{i=1}^k p_i \log_2 p_i$$

p_i : proportion of the samples that belongs to class c for a particular node.

*This is the definition of entropy for all non-empty classes ($p \neq 0$). The entropy is 0 if all samples at a node belong to the same class.



When to stop splitting?

Definition of Decision Tree:- By default, DT uses all feature in model and creates a overfitting model.

As a problem usually has a large set of features, it results in large number of split, which in turn gives a huge tree. Such trees are *complex and can lead to overfitting*.

$f_1, f_2, f_3, f_4, \dots, f_{10}$

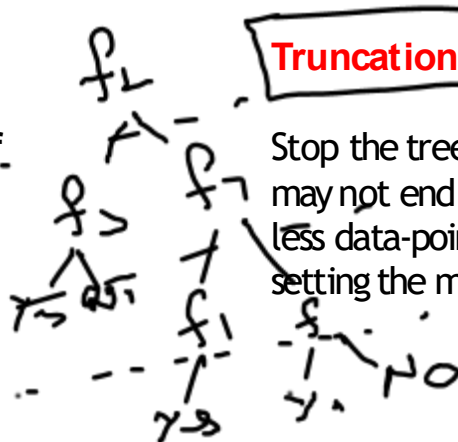
Pruning

Remove the **branches that make use of features having low importance**. This way, we reduce the complexity of tree, and thus increasing its predictive power by reducing overfitting.



Truncation

Stop the tree while it is still growing so that it may not end up with leaves containing very less data-points, one way to do this is by setting the minimum number

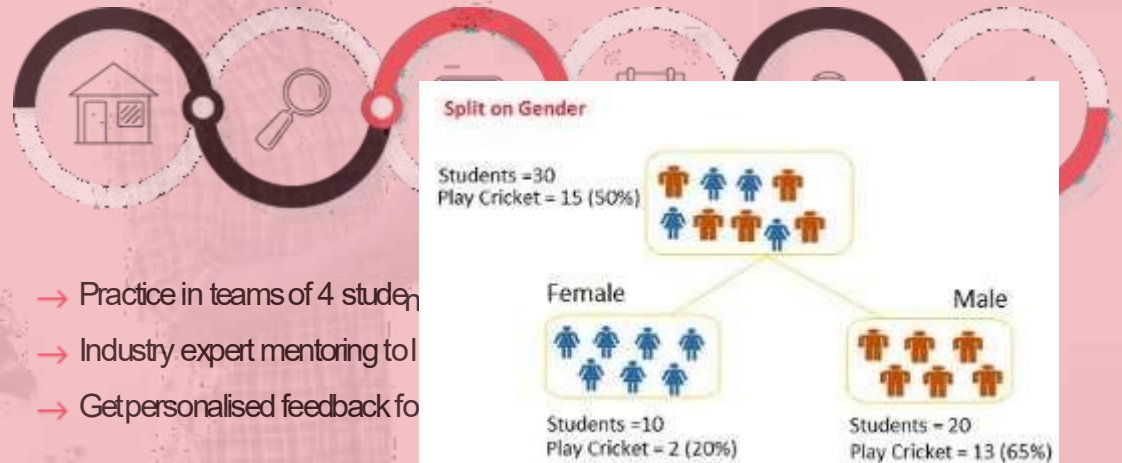


instead of grow tree fully.

Q4badd Title

Question: we want to segregate the students based on target variable (playing cricket or not). we split the population using input variables Gender find the homogeneity of sub-nodes using Gini.

- A) 0.51
- B) 0.59
- C) 0.33
- D) 0.49

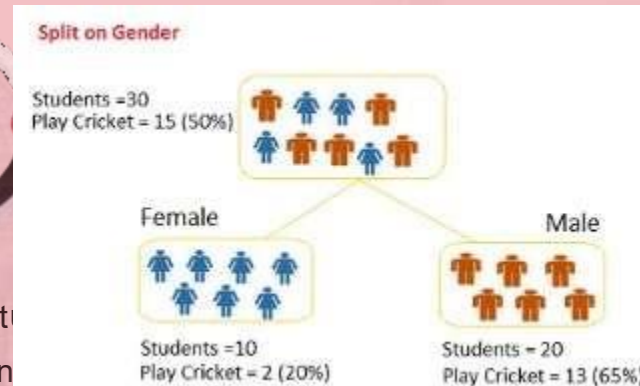


- Practice in teams of 4 students
- Industry expert mentoring to learn
- Get personalised feedback for

Q4 (No Solution)

Question: we want to segregate the students based on target variable (playing cricket or not). we split the population using input variables Gender find the homogeneity of sub-nodes using Gini.

- A) 0.51
- B) 0.59
- C) 0.33
- D) 0.49



→ Practice in teams of 4 students

→ Industry expert mentoring

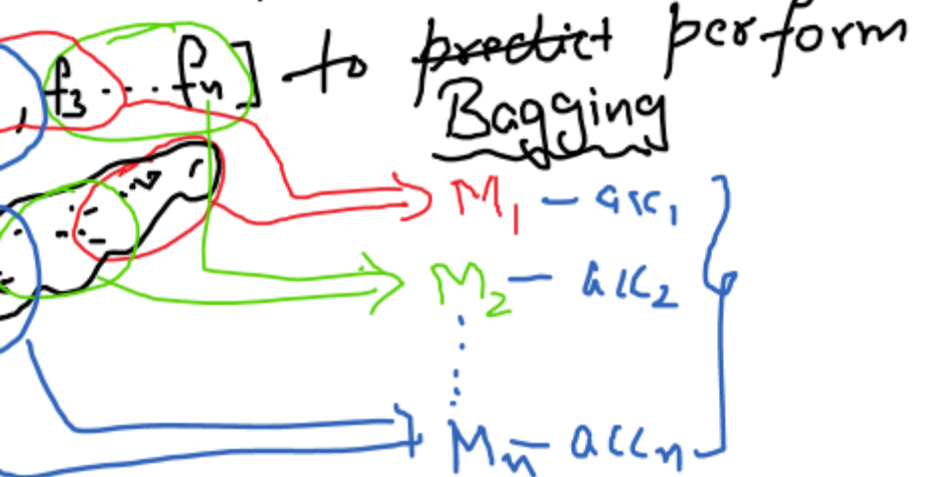
→ Get personalised feedback for improvements

Calculate, Gini for sub-node Female = $(0.2) \times (0.2) + (0.8) \times (0.8) = 0.68$

Gini for sub-node Male = $(0.65) \times (0.65) + (0.35) \times (0.35) = 0.55$

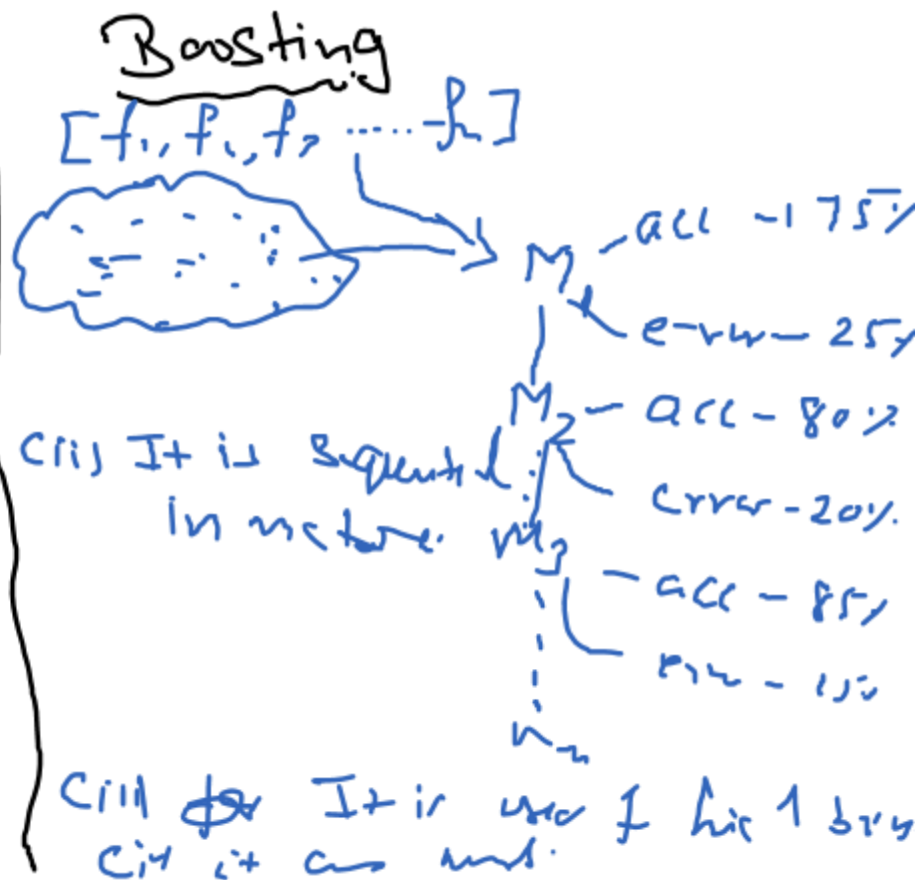
Calculate weighted Gini for Split Gender = $(10/30) \times 0.68 + (20/30) \times 0.55 = 0.59$

Ensemble Learning → we build multiple model and use these model collectively. edGrad



$$Acc = \frac{acc_1 + acc_2 + acc_3 + \dots + acc_n}{n}$$

- is parallel in nature
 + is used for High Variance
 + helps to more stable model

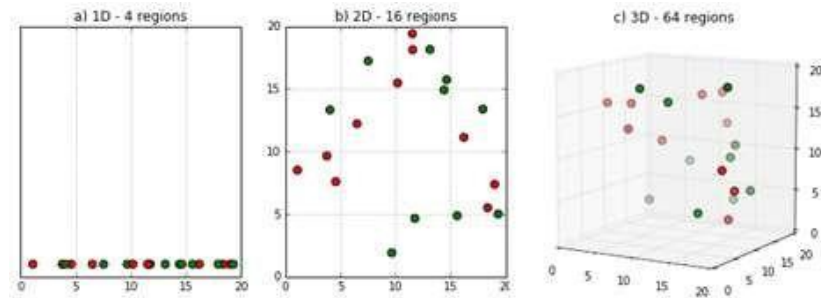


What is the curse of dimensionality?

PCA → Principal Component Analysis

Sparsity of data occurs when moving to higher dimensions. the volume of the space represented grows so quickly that the data cannot keep up and thus becomes sparse

The second issue that arises is related to sorting or classifying the data. In low dimensional spaces, data may seem very similar but the higher the dimension the data points may seem far.



100 features
→ 300-400 dimen.

Infinite Features Requires Infinite Training

x_1, x_2

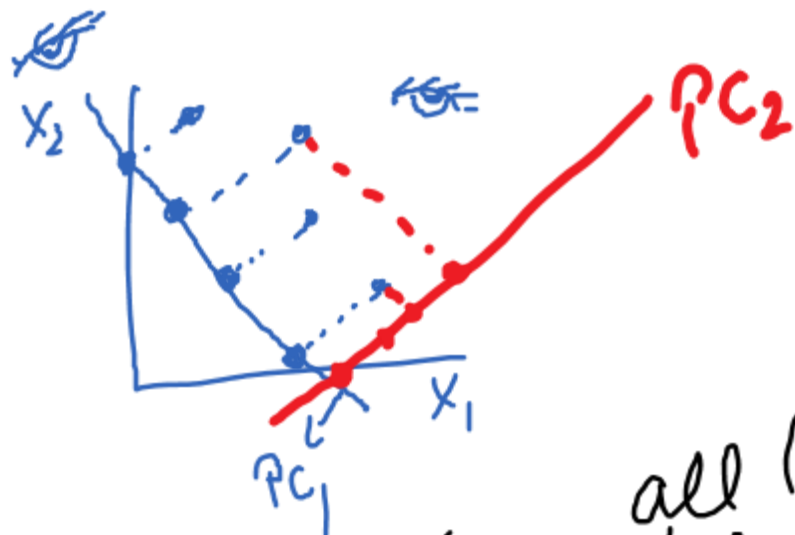
A careful choice of the number of dimensions (features) to be used is the prerogative of the data scientist training the Model

features create overfitting problem.

PCA → reduce

Transfer Data into lower dimen. without

Gen-



Limitations After projecting data to PCA we will get $\downarrow$ there will be no existence of

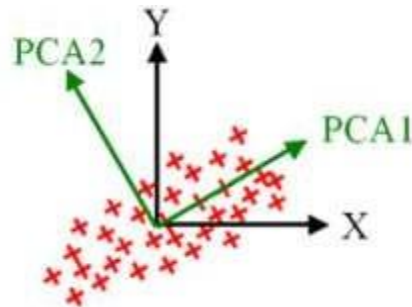


all Principal Components are $\perp$ to each other
 $X_1 \neq PC_1$
 PC is linear combination of features.

$$\alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n$$

$$= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Principal Components Analysis is an unsupervised learning class of statistical techniques used to explain data in high dimension using smaller number of variables called the principal components.

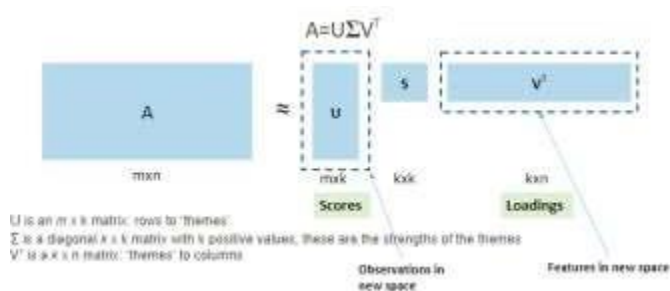


The loading vector Φ_1 with elements $\Phi_{11}, \Phi_{21}, \dots, \Phi_{p1}$ defines a direction in the feature space along which there is maximum variance in the data.

Thus, if we are to project the n data points $x_1, x_2, \dots, x_n$ onto this direction, then projected values are the actual principal component scores $z_{11}, z_{21}, \dots, z_{n1}$.

After the first principal components, Z_1 of the features has been determined, then the second principal component is the linear combination of $X_1, X_2, \dots, X_p$ that has the highest variance out of all the linear combinations that are uncorrelated with Z_1 . The second principal component scores $z_{12}, z_{22}, \dots, z_{n2}$ take the form

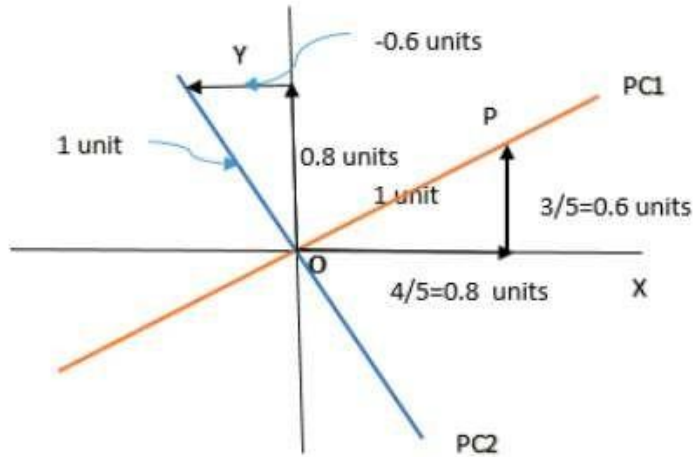
PCA is useful when you have a large number of **potentially correlated variables**. Also, PCA can be used to visualize complex datasets.



We saw that if each row in the original dataset A ($m \times n$) represents a user and each column an item such as a book, then:

- The matrix U ($m \times k$) maps the users to the new k themes/latent variables
- The diagonal matrix S ($k \times k$) represents the strength of each feature
- Matrix V' ($k \times n$) maps each of the k themes/latent variables to the original n features.

Q. Add Title

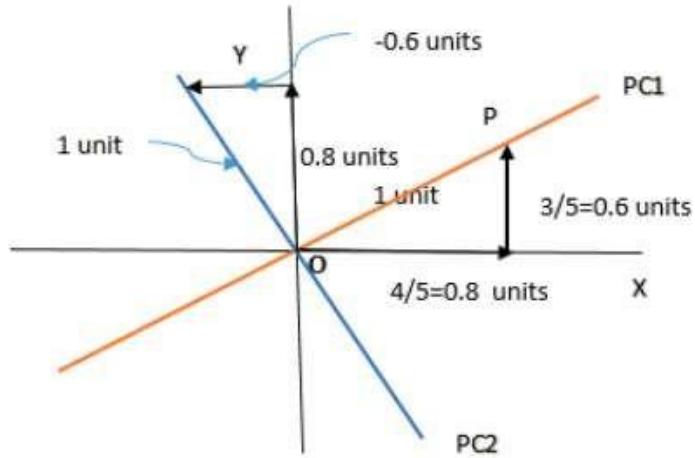


What are the PC1 and PC2 values for given $x=3$ and $y=2$?

- A) 3.4 & 1.2
- B) 3.6 & -0.2**
- C) 3.4 & -0.2
- D) 3.6 & 1.2

→ Get personalised feedback for improvements

Q15) Answer & Title



What are the PC1 and PC2 values for given $x=3$ and $y=2$?

- A) 3.4 & 1.2
- B) 3.6 & -0.2**
- C) 3.4 & -0.2
- D) 3.6 & 1.2

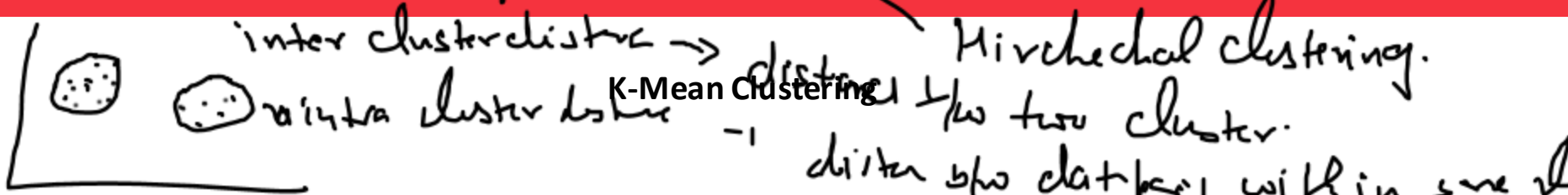
→ Get personalised feedback for improvements

CLUSTERING

→ Unsupervised ML

K Mean Clustering

UpGrad



- The number of clusters that you want to divide your data points into, i.e. the value of K has to be pre-determined.
- The choice of the initial cluster centers can have an impact on the final cluster formation.
- The clustering process is very sensitive to the presence of outliers in the data.
- Since the distance metric used in the clustering process is the Euclidean distance, you need to bring all your attributes on the same scale. This can be achieved through standardization.
- The K-Means algorithm does not work with categorical data.

Inter cluster is
Intra cluster is

Choosing the number of clusters K

Elbow method

- Compute clustering algorithm (e.g., k-means clustering) for different values of k. For instance, by varying k from 1 to 10 clusters.
- For each k, calculate the total within-cluster sum of square (wss)
- Plot the curve of wss according to the number of clusters k.
- The location of a bend (knee) in the plot is generally considered as an indicator of the appropriate number of clusters.

CLUSTERING

Hierarchical Clustering algorithm

Given a set of N items to be clustered, the steps in the hierarchical clustering are:

- Calculate the $N \times N$ distance (similarity) matrix, which calculates the distance of each data point from the other.
- Start by assigning each item to its own cluster, so that if you have N items, you now have N clusters, each containing just one item.
- Find the closest (most similar) pair of clusters and merge them into a single cluster, so that now you have one less cluster.
- Compute distances (similarities) between the new cluster and each of the old clusters.
- Repeat steps 3 and 4 until all items are clustered into a single cluster of size N .

Single Linkage

Here, the distance between 2 clusters is defined as the shortest distance between points in the two clusters

Complete Linkage

Here, the distance between 2 clusters is defined as the maximum distance between any 2 points in the clusters

Average Linkage

Here, the distance between 2 clusters is defined as the average distance between every point of one cluster to every other point of the other cluster.

① Assume each point as separate cluster.

2. Find shortest dist b/w points and create

each into a single cluster

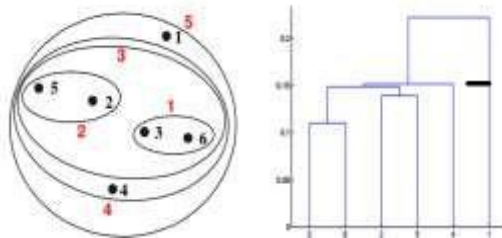
We don't need to start with very large cluster value of 11.

Popkto add Title

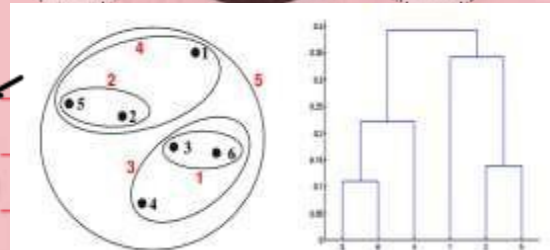
Which of the following clustering representations and dendrogram depicts the use of MIN or Single link proximity function in hierarchical clustering?

	p1	p2	p3	p4	p5	p6
p1	0.0000	0.2357	0.2218	0.3688	0.3421	0.2347
p2	0.2357	0.0000	0.1483	0.2042	0.1388	0.2540
p3	0.2218	0.1483	0.0000	0.1513	0.2843	0.1100
p4	0.3688	0.2042	0.1513	0.0000	0.2932	0.2216
p5	0.3421	0.1388	0.2843	0.2932	0.0000	0.3921
p6	0.2347	0.2540	0.1100	0.2216	0.3921	0.0000

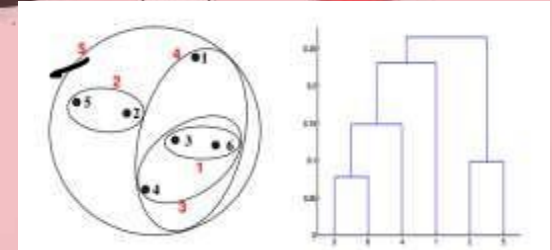
Distance Matrix



(A)



(B)



(C)

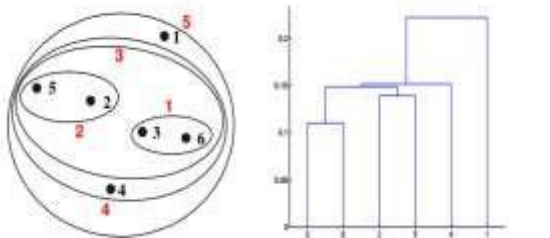
Pop 6 (Answer)

Q. 6 (Answer)

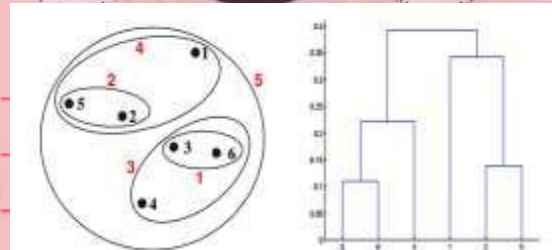
Which of the following clustering representations and dendrogram depicts the use of Single link proximity function in hierarchical clustering?

	p1	p2	p3	p4	p5	p6
p1	0.0000	0.2357	0.2218	0.3688	0.3421	0.2347
p2	0.2357	0.0000	0.1483	0.2042	0.1388	0.2540
p3	0.2218	0.1483	0.0000	0.1513	0.2843	0.1100
p4	0.3688	0.2042	0.1513	0.0000	0.2932	0.2216
p5	0.3421	0.1388	0.2843	0.2932	0.0000	0.3921
p6	0.2347	0.2540	0.1100	0.2216	0.3921	0.0000

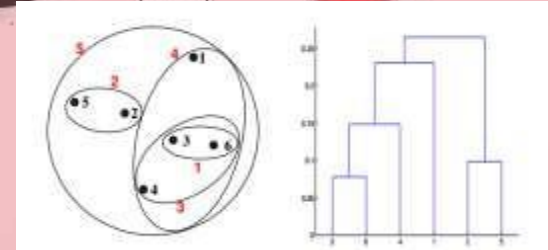
Distance Matrix



(A)

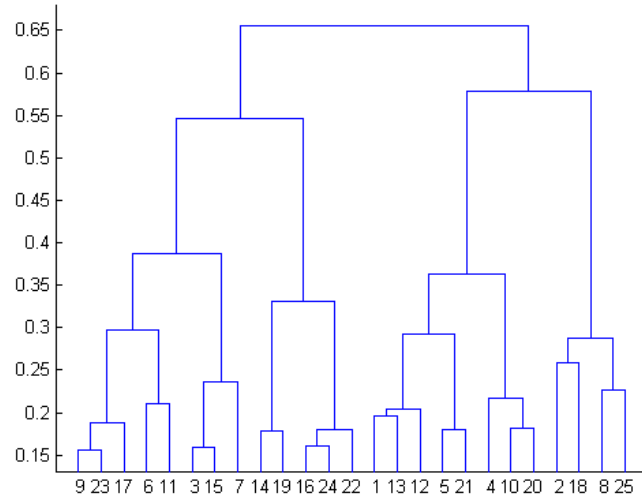


(B)



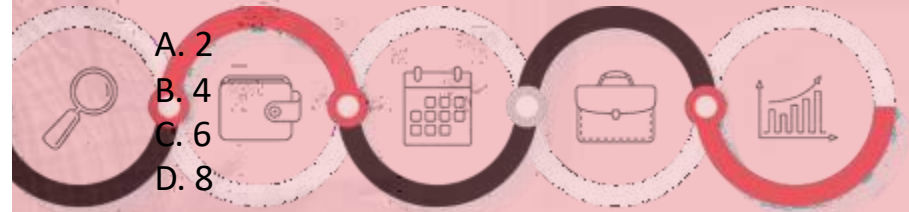
(C)

Q7badd Title



What is the most appropriate no. of clusters for the data points represented by the following dendrogram?

- A. 2
- B. 4
- C. 6
- D. 8

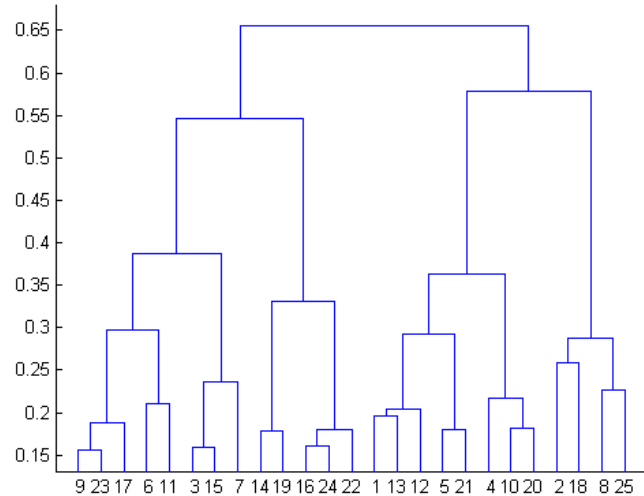


Teams of 4 students

Expert mentoring to learn better

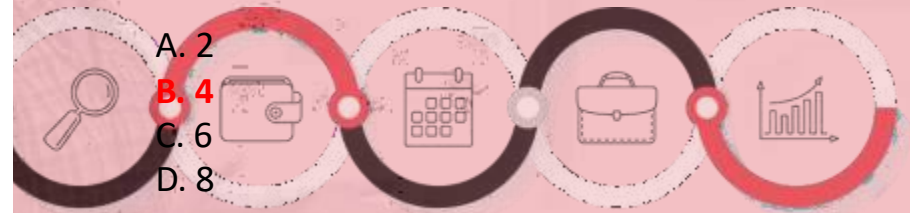
Personalised feedback for improvements

Qn 7: Answer & Title



What is the most appropriate no. of clusters for the data points represented by the following dendrogram?

- A. 2
- B. 4**
- C. 6
- D. 8

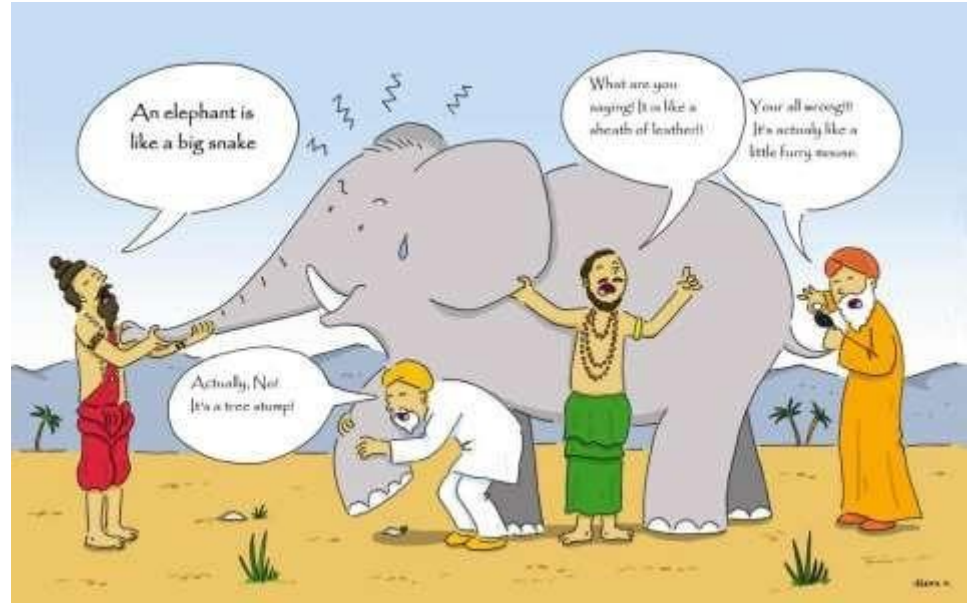


Teams of 4 students

Expert mentoring to learn better

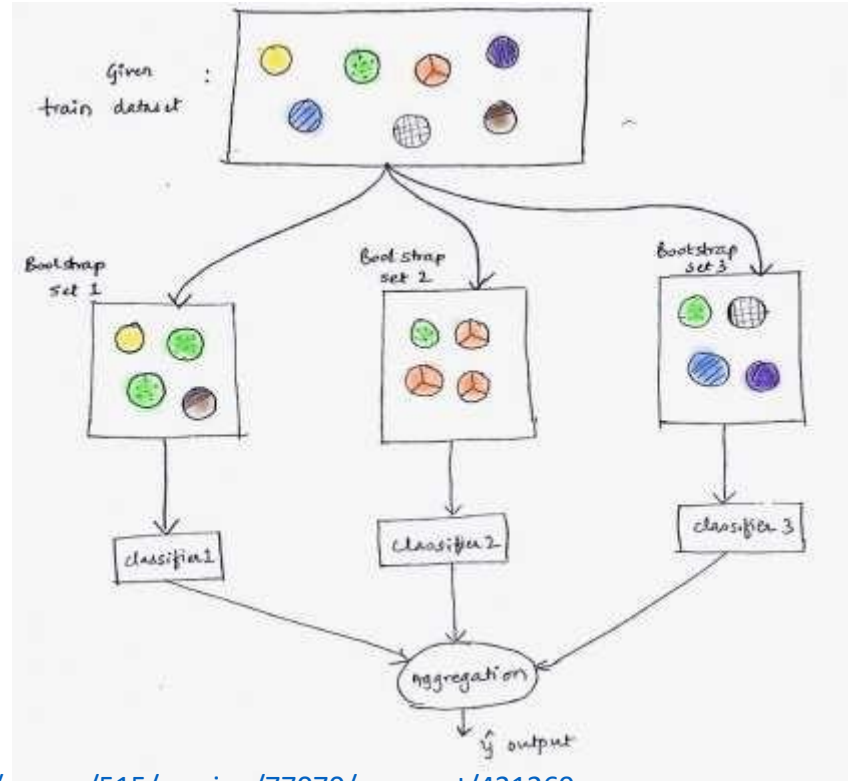
Personalised feedback for improvements

Ensemble learning is a machine learning paradigm where multiple models (often called “weak learners”) are trained to solve the same problem and combined to get better results.



<https://towardsdatascience.com/ensemble-methods-bagging-boosting-and-stacking-c9214a10a205>

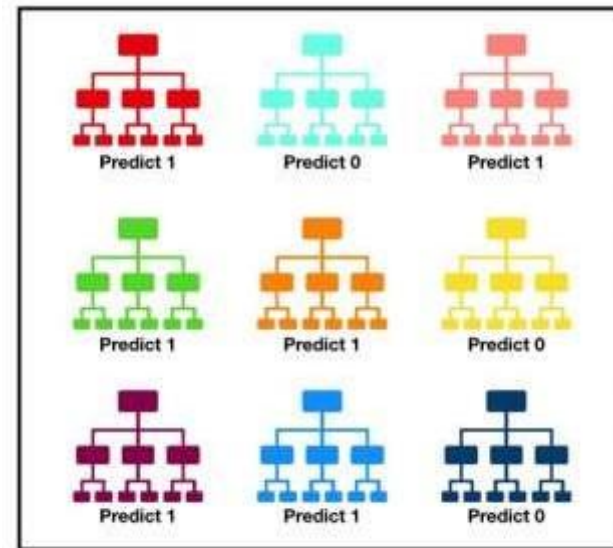
Bagging, that often considers homogeneous weak learners, learns them independently from each other in parallel and combines them following some kind of deterministic averaging process



<https://learn.upgrad.com/v/course/515/session/77070/segment/431269>

Random forest, like its name implies, consists of a large number of individual decision trees that operate as an ensemble. Each individual tree in the random forest spits out a class prediction and the class with the most votes becomes our model's prediction

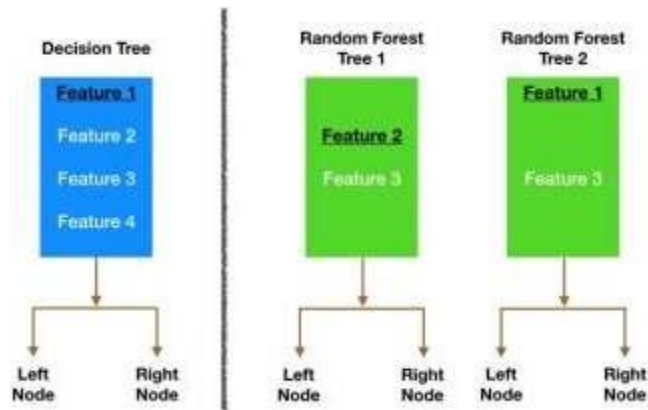
The reason for this wonderful effect is that the trees protect each other from their individual errors (as long as they don't constantly all err in the same direction)

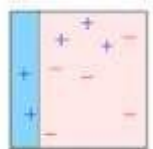


Tally: Six 1s and Three 0s
Prediction: 1

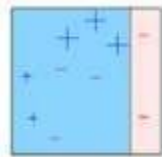
Bagging (Bootstrap Aggregation) — Decision trees are very sensitive to the data they are trained on — small changes to the training set can result in significantly different tree structures. Random forest takes advantage of this by allowing each individual tree to randomly sample from the dataset with replacement, resulting in different trees.

Feature Randomness — In a normal decision tree, while splitting a node, we consider all the features and pick the one that produces the most separation. In contrast, each tree in a random forest can pick only from a random subset of features. This forces even more variation amongst the trees in the model and ultimately results in lower correlation across trees and more diversification.

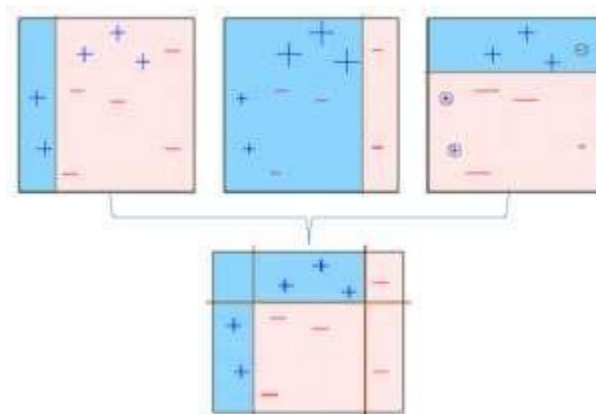




Base model is created on a subset

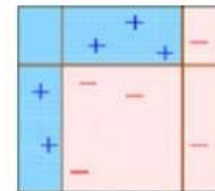


The observations which are incorrectly predicted, are given higher weights and an another model is created and predictions are made on the dataset



multiple models are created, each correcting the errors of the previous model.

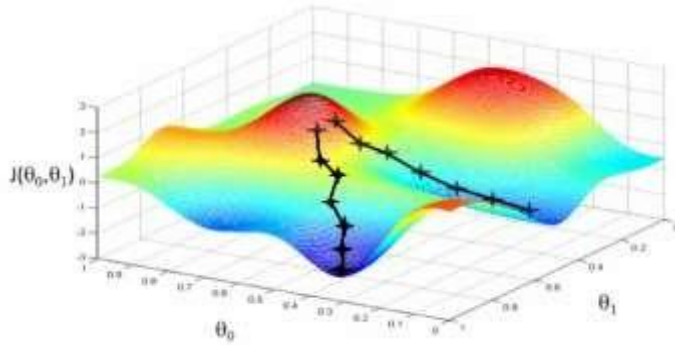
The final model (strong learner)



. The individual models would not perform well on the entire dataset, but they work well for some part of the dataset. Thus, each model actually boosts the performance of the ensemble.

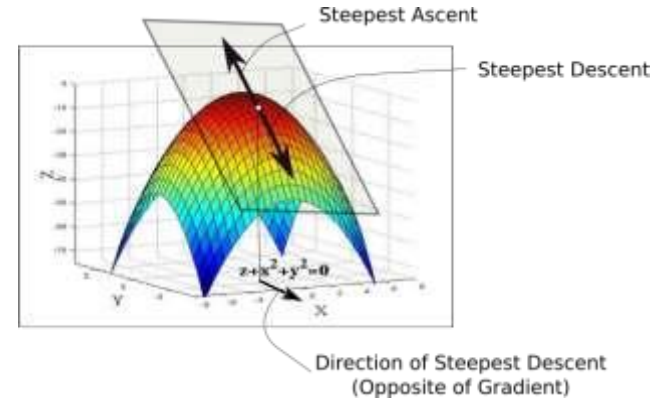
Boosting, that often considers homogeneous weak learners, learns them sequentially in a very adaptive way (a base model depends on the previous ones) and combines them following a deterministic strategy

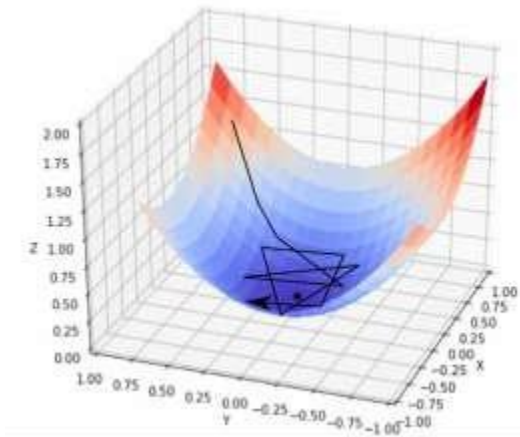
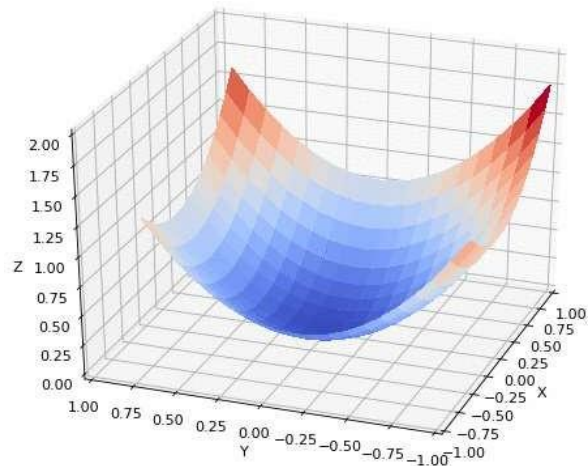
<https://medium.com/mlreview/gradient-boosting-from-scratch-1e317ae4587d>



Gradient descent is an optimization algorithm used to minimize some function by iteratively moving in the direction of steepest descent as defined by the negative of the gradient

While the direction of the gradient tells us which direction has the steepest ascent, it's magnitude tells us how steep the steepest ascent/descent is. So, at the minima, where the contour is almost flat, you would expect the gradient to be almost zero. In fact, it's precisely zero for the point of minima.





In practice, we might never *exactly* reach the minima, but we keep oscillating in a flat region in close vicinity of the minima. As we oscillate over this region, the loss is almost the minimum we can achieve, and doesn't change much as we just keep bouncing around the actual minimum. Often, we stop our iterations when the loss values haven't improved in a pre-decided number, say, 10, or 20 iterations. When such a thing happens, we say our training has converged, or convergence has taken place.



Thank You!